Sabyasachi Mukherjee, M.Sc.

Email: smukherj(at)smail(dot)uni-koeln.de Website: https://www.mi.uni-koeln.de/ smukherj/

Interests

Sphere packings, discrete geometry

Education

PhD in Mathematics April 2026 - Present
Universität zu Köln Cologne, Germany

M.Sc. in Mathematics March 2022
Georg-August-Universität Göttingen Göttingen, Germany
Thesis: *The Möbius Function on Arithmetic Progressions*

B.S. in Mathematics May 2017
Shiv Nadar University Noida, India

Teaching & Grading Experience

Tutor Georg-August-Universität Göttingen
Real Analysis I Winter 2020

Tutor Georg-August-Universität Göttingen
Functional Analysis Summer 2020

Tutor Georg-August-Universität Göttingen
Real Analysis I Winter 2019

Grader Georg-August-Universität Göttingen
Linear Algebra and Analytic Geometry Summer 2019

Selected Talks

Hasse Principle for Non-singular Quartic Hypersurfaces February 2022
Analytic Number Theory Working Seminar

A Sharp Version of Halász's Theorem December 2020
Göttingen Number Theory Working Seminar

Large subsets of $\mathbb{F}_q^n$ without three-term arithmetic progressions July 2020
Göttingen Number Theory Oberseminar

Conferences & Summer Schools (Previous and Upcoming)

Serre 100 (Conference in Honour of Jean-Pierre Serre) September 15 – September 16, 2026
Paris, France

Quadratic Lattices in Prague 2026 August 17 – August 21, 2026
Prague, Czechia

Random Geometry and Number Theory June 22 – July 3, 2026
Lausanne, Switzland (online)

Summer School in Analytic Number Theory June 28 – July 2, 2021
Paris, France

Hausdorff Centre of Mathematics: "The Circle Method" May 10–21, 2021
Bonn, Germany

Technical Skills and Languages

Python, MATLAB, LaTeX, English (C2), German (C1), Bengali (C2), Hindi (C2)

Work Experience

Industry: Data Science 2022 – 2025